Estimating the Baseline and Crisis Windows

Main specification design

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1 Question and design

Question. What determines the training decision in a frictional, skill-segmented labour market, and how does it respond to aggregate shocks over time? The financial crisis and COVID both raised unemployment but had opposite effects on vacancies. Two questions follow: which parameters shift during a crisis, and how does the economy adjust between the resulting equilibria?

Premise. Crises are not the same repeating fluctuation of a business cycle but qualitatively distinct events, so a crisis is a shift between steady states rather than a displacement within one.

Design. The two baselines are one economy observed twelve years apart, and I estimate them jointly. On a balanced growth path a parameter keeps one value in units of output, so it takes its own value in each baseline only where what it describes changed between them. A crisis is too short to change the population and its technologies, which keep their baseline values; the conditions that can change within months (demand, unemployment income, job destruction and the cost of creating a job) are estimated, the same ones in both crises. Each window is a steady state.

2 The baselines

The joint objective is the sum of the objectives of 2003–07 and 2015–19, each with its own data moments and weights. Table 1 gives the reason for each parameter, and Table 2 the changes in the data that require a parameter to take its own value.

Table 1: Parameters in the Joint Estimation of the Two Baselines

Parameter	Baselines	Reason
Ability correlation, ρ_x	common	Aptitudes are fixed at birth, and their joint distribution is a property of the population.
Training cost, c	own value	Largely a psychological cost of studying (Cunha, Heckman and Navarro 2005), which changes over decades as people's attitudes to education change.
General productivity, A	own value	Growth.
Unemployment income, b_U, b_S	common	Moves roughly in proportion to productivity (Chodorow-Reich and Karabarbounis 2016), so it keeps one value in units of output; government policy changes it only temporarily, in crises.
Skilled productivity, P_S	own value	The demand for skilled work changes over decades (Goldin and Katz 2008; Beaudry, Green and Sand 2016).
Quality of unskilled redraws, α_U [a]	own value	Pinned by the share of separations in turnover; the SIPP measures within-job wage changes differently after its 2014 redesign.
Matching efficiency, μ_U, μ_S	own value	Fell after the Great Recession and stayed low through the 2010s (Barlevy et al. 2024).
Quality shock rate, λ_U, λ_S	own value	Pinned by the sum of the separation and wage-change rates, as for α_U.
Cost of search on the job, σ_S	common	OJS cost keeps one value in units of output.
Vacancy cost, k_U, k_S	own value	Depends on how firms recruit: recruiting intensity per vacancy fell after 2007 (Davis, Faberman and Haltiwanger 2013), one reason vacancies yielded fewer hires through the 2010s (Barlevy et al. 2024).

NOTE.—Common: one value for 2003–07 and 2015–19. Own value: a value for each window.
[a]When a quality shock hits an unskilled match, its new quality p is drawn from the distribution $G(p) = p^{\alpha_U}$ on $[0, 1]$, with density $g(p) = \alpha_U\, p^{\alpha_U - 1}$.

Table 2: The Two Baseline Windows in the Data

	2003–07	2015–19	Change (%)	Implication
Unskilled market:				
Tightness	0.449	0.778	+73.4	μ_U, k_U own value
Job-finding rate	0.301	0.293	−2.7	
Separation rate	0.015	0.013	−12.9	α_U, λ_U own value
Wage-change rate	0.151	0.208	+38.5	
Unemployment rate	0.064	0.056	−11.5	
Mean log wage	6.530	6.535	+0.5	
Skilled market:				
Tightness	1.163	1.805	+55.2	μ_S, k_S own value
Job-finding rate	0.319	0.322	+0.7	
Separation rate	0.006	0.006	−5.2	λ_S own value
Wage-change rate	0.129	0.222	+72.2	
Job-to-job rate	0.008	0.009	+6.8	
Unemployment rate	0.026	0.025	−5.2	
Mean log wage	6.915	6.920	+0.5	
Composition:				
Training share	0.044	0.052	+18.5	c own value
Skilled share	0.302	0.377	+24.9	

NOTE.—Monthly rates. Wages are real weekly earnings; their change is the difference in mean log wages times 100. The wage-change rates come from the SIPP: from the hourly rate with a break filter in 2003–07, and from earnings in 2015–19, after its 2014 redesign. Implication: the treatment in Table 1 that the bracketed rows require together. Tightness rose while job finding barely moved, so each vacancy yields fewer hires: matching efficiency differs, and free entry then pins a different vacancy cost. Blank: the treatment rests on the reason in Table 1.

3 The crisis windows

Each crisis window is a steady state, estimated with every held parameter at its value in the preceding baseline; the transition between the two steady states answers the second question. The same parameters are estimated in both crises (Table 3): general productivity A, unemployment income b_U and b_S, the redraw parameter α_U, the shock rates λ_U and λ_S, and the vacancy costs k_U and k_S. Measured turnover pins the shock rates almost one for one, and the tightness pair pins the vacancy costs through free entry. Which of these parameters moved in each crisis, and by how much, is the result.

Table 3: Parameters in the Crisis Windows

Parameter	In a crisis	Reason
Ability correlation, ρ_x	held	Aptitudes are fixed at birth.
Training cost, c	held	A psychological cost that changes over decades (Table 1).
General productivity, A	estimated	Demand can change within months: the fall in household net worth cut demand and employment in the financial crisis (Mian and Sufi 2014), and output collapsed and recovered within the COVID window.
Unemployment income, b_U, b_S	estimated	Government policy can change it within months, through UI benefits or other relief, as in both crises (Rothstein 2011; Ganong, Noel and Vavra 2020).
Skilled productivity, P_S	held	The demand for skilled work changes over decades (Table 1).
Quality of unskilled redraws, α_U	estimated	Job destruction can change within months: the dispersion of shocks to producers rises sharply in recessions (Bloom et al. 2018), and in both crises job losses fell most on less-educated workers (Hoynes, Miller and Schaller 2012; Mongey, Pilossoph and Weinberg 2021).
Matching efficiency, μ_U, μ_S	held	Matching technology changes over decades (Table 1). In the financial crisis job finding fell as much as the fall in tightness implies; in COVID it rose by more, during the rehiring of laid-off workers (Barlevy et al. 2024), which needs no change in matching technology.
Quality shock rate, λ_U, λ_S	estimated	Part of the shock process, as α_U.
Cost of search on the job, σ_S	held	A search technology, which does not change within months.
Vacancy cost, k_U, k_S	estimated	The cost of creating a job can change within months: credit frictions add to it (Wasmer and Weil 2004), and after the Lehman bankruptcy firms whose lenders were weaker cut employment more (Chodorow-Reich 2014).

NOTE.—Held: at the value of the preceding baseline. Estimated: on each crisis window, the same parameters in both crises.

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